

Cellphone Price Range Classification

Predicting phone price category from hardware specifications

1. Project Overview

This report summarizes an end-to-end machine learning workflow that classifies mobile phones into one of four price categories using 20 hardware and connectivity specifications (battery power, RAM, camera resolution, screen size, processor details, and connectivity features such as Bluetooth, dual-SIM, 3G/4G, and Wi-Fi). The dataset contains 2,000 phones with no missing values and no duplicate records.

Target Variable

Value	Price Range	Count
0	Low cost	500
1	Medium cost	500
2	High cost	500
3	Very high cost	500

The classes are perfectly balanced (500 phones each), so accuracy is a reliable performance metric and no class-imbalance correction is needed.

Data Quality Checks

- Missing values: 0 across all 21 columns
- Duplicate rows: 0
- Outliers (IQR method): negligible — 18 in front-camera megapixels (fc), 2 in pixel height (px_height); all other numerical features had 0 outliers

2. Exploratory Data Analysis

2.1 Target Distribution

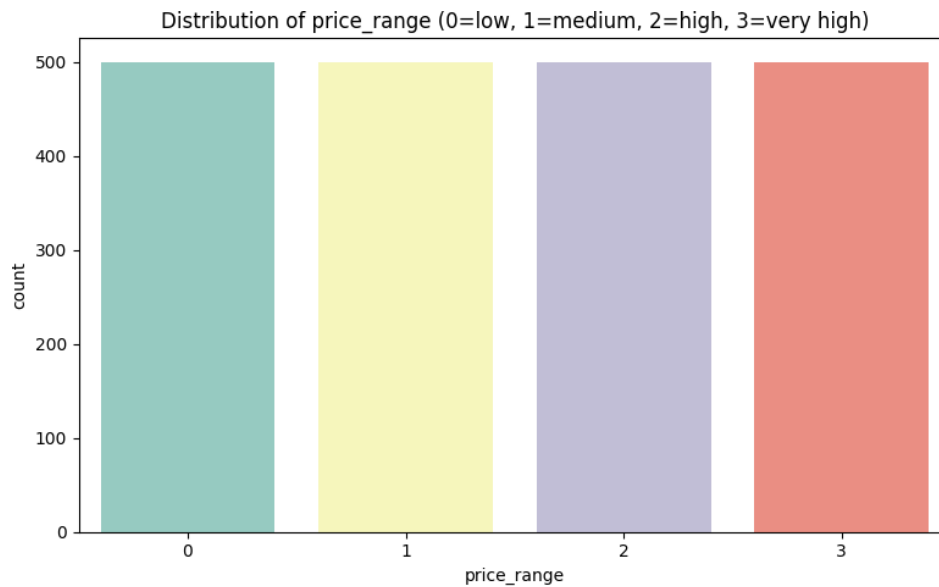


Figure 1. The four price-range classes are evenly distributed (500 phones each).

2.2 Numerical Feature Distributions

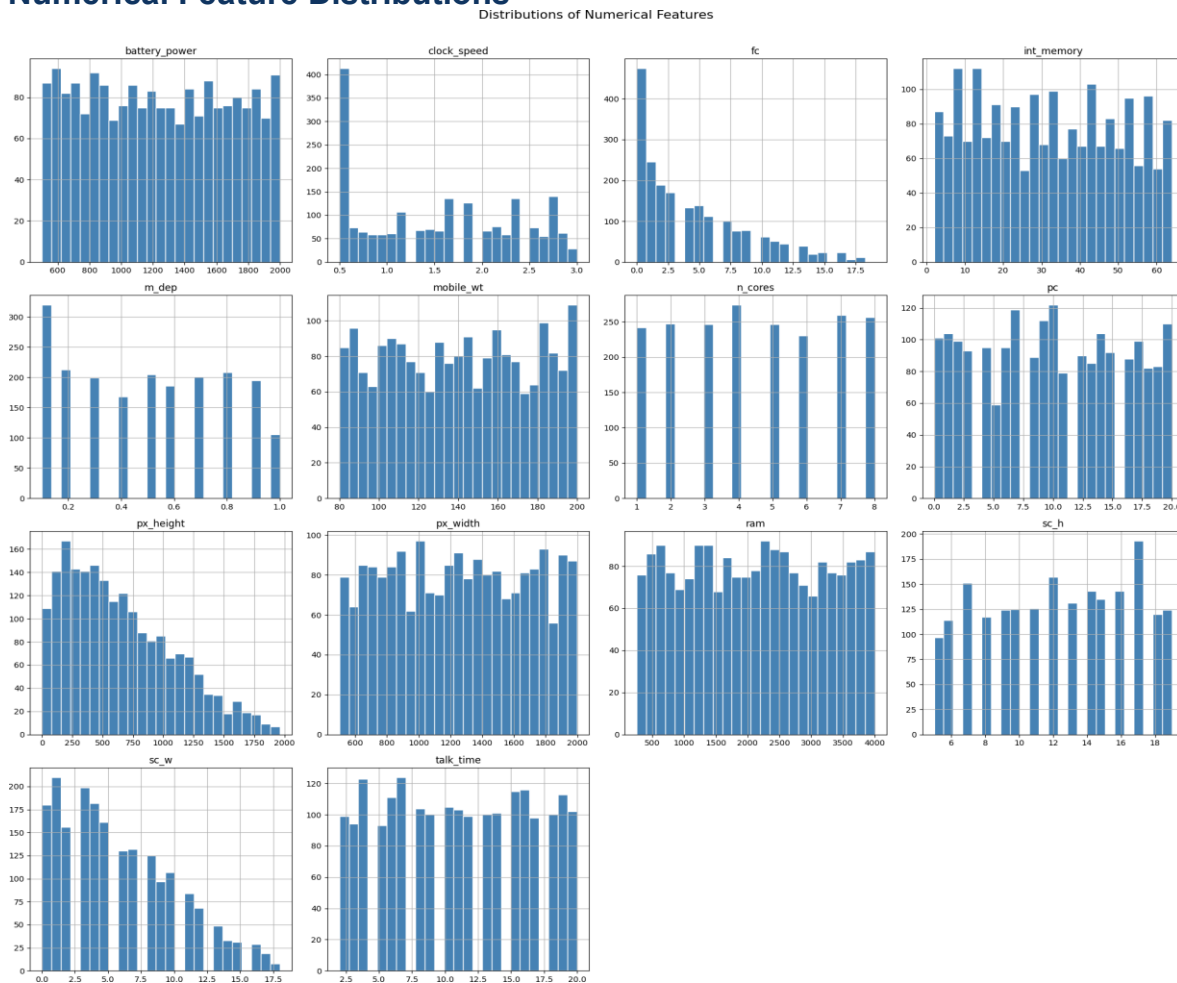


Figure 2. Histograms of the 14 numerical features. Most features (battery power, RAM, pixel dimensions, talk time) are close to uniformly distributed, consistent with a synthetically generated specification dataset.



Figure 3. Boxplots of numerical features confirming the IQR outlier scan — only front camera megapixels and pixel height show a small number of outlying points.

2.3 Categorical Feature Distributions

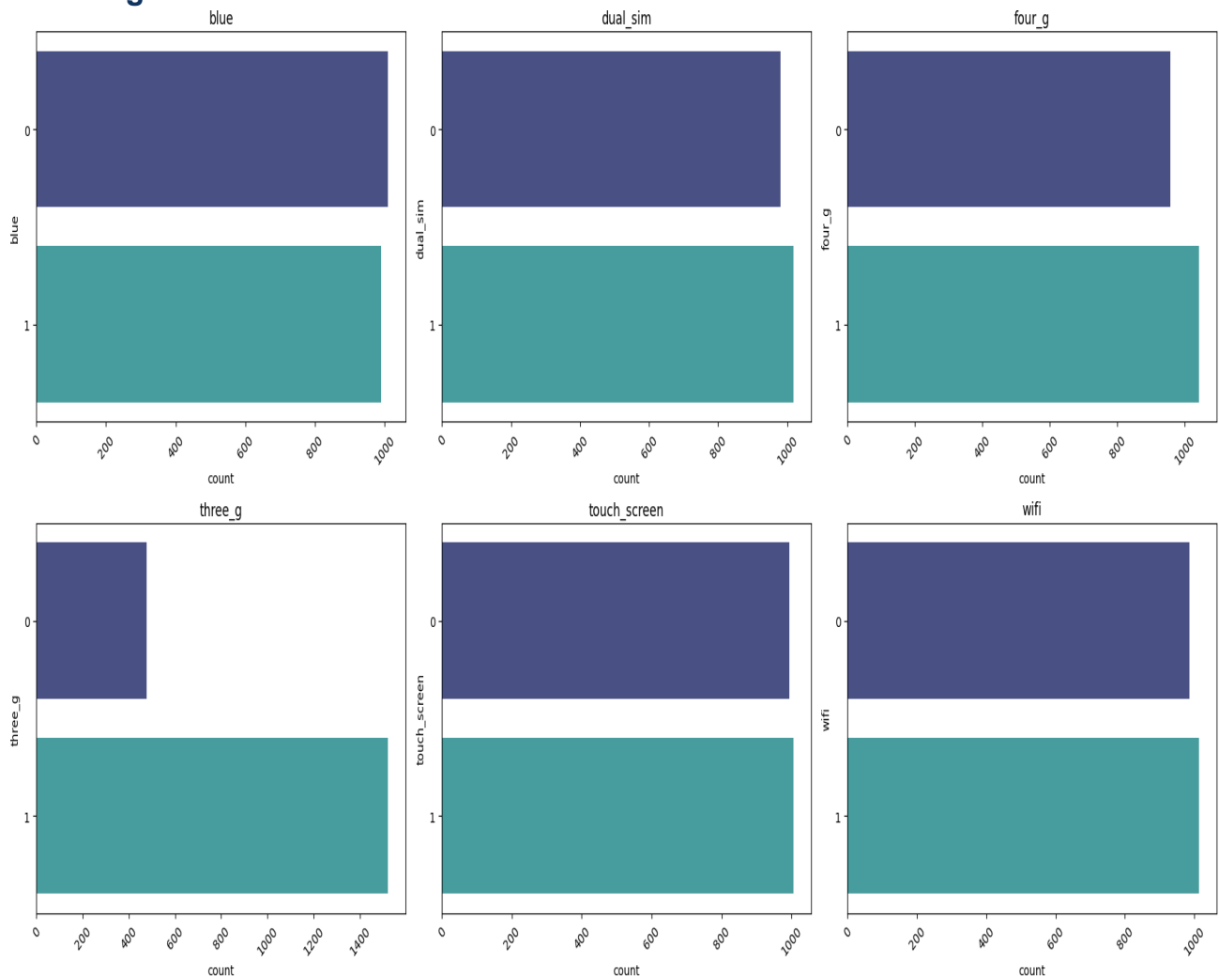
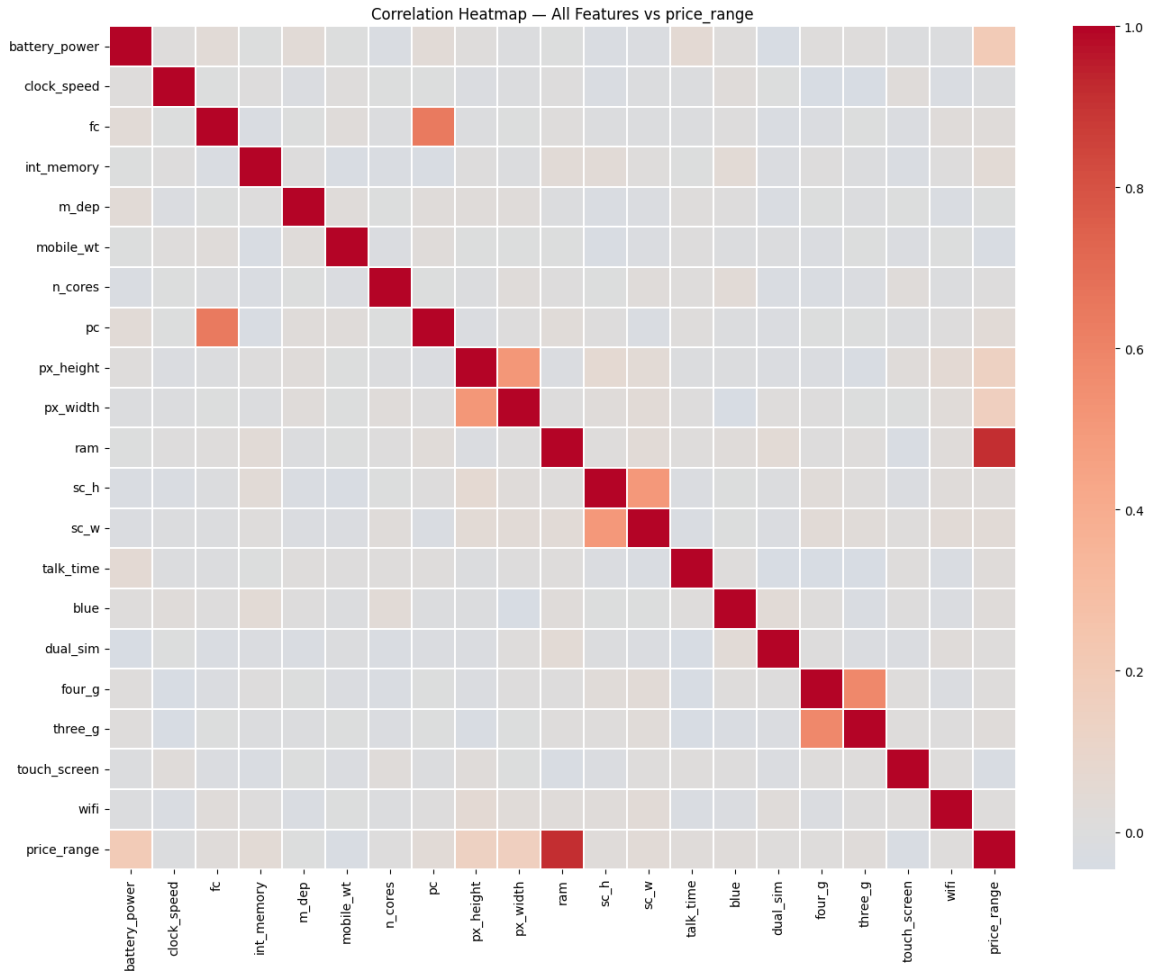


Figure 4. Binary connectivity/hardware flags (Bluetooth, dual-SIM, 4G, 3G, touch screen, Wi-Fi) are roughly balanced between 0/1 across the dataset.

2.4 Correlation with Price Range



Feature	Correlation with price_range
ram	0.917
battery_power	0.201
px_width	0.166
px_height	0.149
int_memory	0.044
All other features	< 0.04 (negligible)

RAM stands out as by far the strongest predictor of price range ($r = 0.92$), followed distantly by battery power and screen resolution (pixel width/height). Most other features — including all the binary connectivity flags — show negligible linear correlation with price.

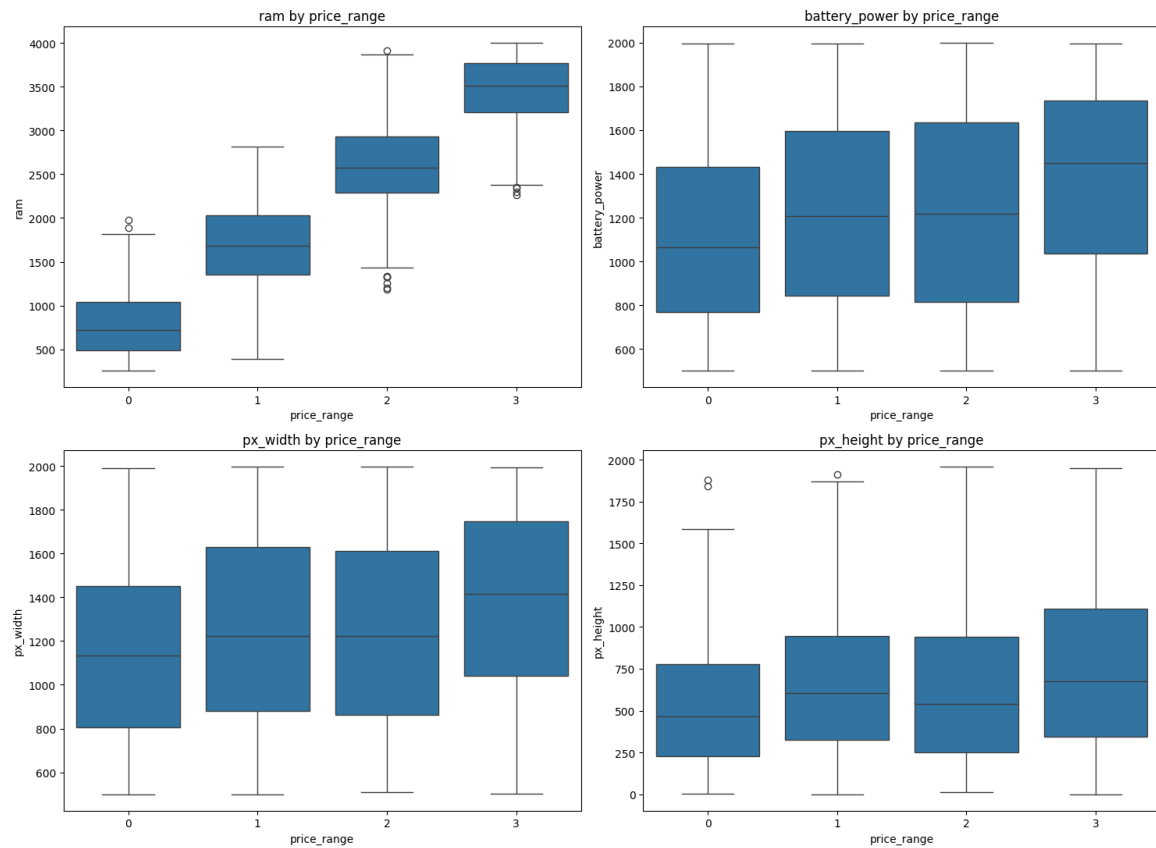


Figure 6. Distribution of the four most predictive numerical features across each price range. RAM shows a clean, near-monotonic separation between classes; the other three overlap considerably.

3. Modeling

3.1 Train/Test Split & Preprocessing

- 80/20 train-test split (1,600 training / 400 test phones), stratified by price_range
- Features standardized with StandardScaler (fit on training data only)

3.2 Models Compared

Nine classifiers were trained and evaluated with 5-fold stratified cross-validation on the training set, then scored on the held-out test set: Logistic Regression, K-Nearest Neighbors, Linear SVM, RBF SVM, Decision Tree, Random Forest, Gradient Boosting, Naive Bayes, and XGBoost.

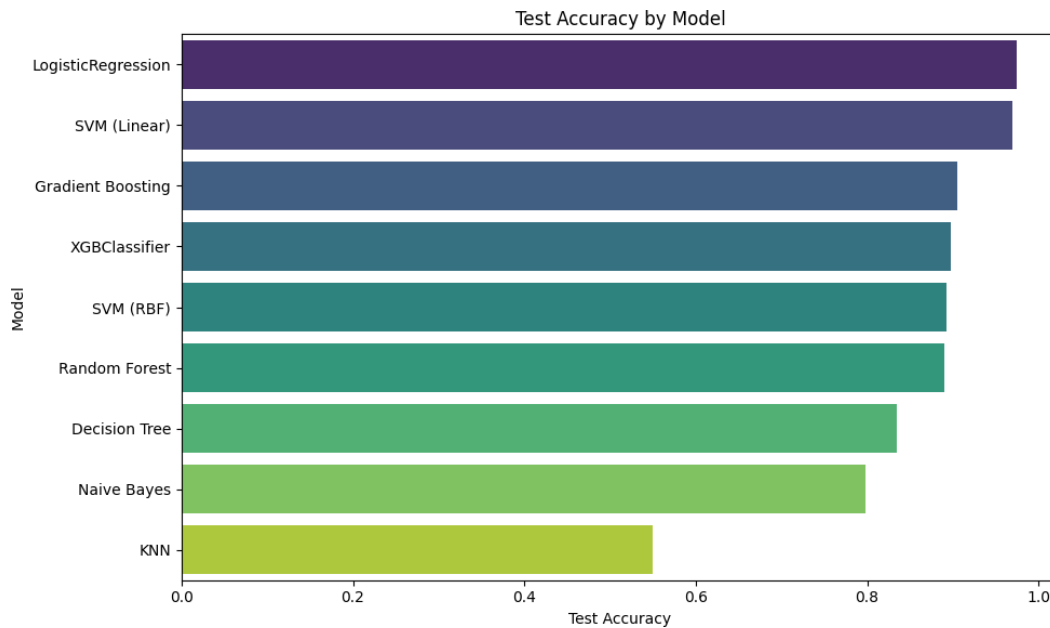


Figure 7. Test accuracy by model. Logistic Regression and the SVM variants lead the field, reflecting that the classes are largely linearly separable once RAM (the dominant feature) is standardized.

Best baseline model: Logistic Regression, with a test accuracy of **97.5%**.

3.3 Confusion Matrix — Logistic Regression (baseline)

	Pred. 0	Pred. 1	Pred. 2	Pred. 3
Actual 0	101	4	0	0
Actual 1	0	91	0	0
Actual 2	0	2	87	3
Actual 3	0	0	1	111

Nearly all misclassifications occur between adjacent price classes (e.g. class 0 confused with class 1, class 2 confused with class 3), which is expected given that price range is an ordinal quantity and boundary cases naturally resemble their neighbors.

3.4 Hyperparameter Tuning

A grid search (5-fold stratified CV) was run over the regularization strength $C \in \{0.01, 0.1, 1, 10, 100\}$ for the best-performing model, Logistic Regression.

- Best parameters: $C = 100$
- Best cross-validation accuracy: 96.9%
- Final tuned model — test accuracy: 97.5%

Per-class performance (tuned model, test set)

Class	Precision	Recall	F1-score	Support
0 (low cost)	1.00	0.94	0.97	105
1 (medium cost)	0.93	1.00	0.96	91
2 (high cost)	0.99	0.97	0.98	92
3 (very high cost)	0.98	0.99	0.99	112
Accuracy			0.97	400
Macro avg	0.97	0.98	0.97	400
Weighted avg	0.98	0.97	0.98	400

The tuned model was serialized with pickle for downstream deployment/inference.

4. Conclusion

- RAM is overwhelmingly the strongest driver of phone price range, far ahead of battery power and screen resolution; most connectivity features (Bluetooth, Wi-Fi, 3G/4G, dual-SIM) carry little predictive signal on their own.
- The dataset is clean and well-balanced, requiring no imputation, deduplication, or class rebalancing.
- A tuned Logistic Regression classifier ($C = 100$) achieved 97.5% test accuracy, matching or beating considerably more complex models (Random Forest, Gradient Boosting, XGBoost, SVM-RBF) — indicating the class boundaries are close to linear once features are standardized.
- Remaining errors are almost entirely between adjacent price classes, consistent with the ordinal nature of the target and natural boundary ambiguity.